25901440 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY201 - NUMERICAL METHODS FOR COMPUTATIONAL PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. What is meant by back substitution in Gauss Elimination method? [K] / [CO1]

2. What is a simple root and a multiple root? [K] / [CO1]

3. What is step size in interpolation? [K] / [CO1]

4. Sketch a rough straight line fit to the following points. [U] / [CO1]

x	0	1	2
y	3	5	7

5. Compare Newton–Raphson method with the bisection method (any two points). [U] / [CO1]

6. The following table shows the boiling point (B) of a solution at different concentrations (C) [U] / [CO1]

C(%)	5	10	15
B(K)	101.2	102.8	105.1

Find the value of ∇B at the highest concentration.

7. When do you apply Newton's forward interpolation formula? [K] / [CO1]

8. Explain the purpose of a difference table. [U] / [CO2]

9. If the tabulated points are equally spaced such that $x_1 = x_0 + h$, prove that the first divided difference reduces to the forward difference, i.e., $f[x_0, x_1] = \frac{\Delta y_0}{h}$. [A] / [CO2]

10. Establish the relation $\Delta = \nabla E$. [A] / [CO2]

11. State the formula of Euler's method for solving a first-order differential equation. Mention the parameters. [K] / [CO2]

12. Using Taylor's series expansion, show that $\left.\frac{\partial T}{\partial y}\right|_{i,j} \approx \frac{T_{i,j} - T_{i,j-1}}{\Delta y}$, where i and j represent grid points corresponding to x and y. [U] / [CO2]

13. Describe the basic concepts employed in Runge-Kutta methods. [K] / [CO2]

14. Rearrange the following procedural steps of the Trapezoidal Rule: [No need to write the entire statement] [An] / [CO2]

    1. Read the initial limit a, the final limit b and the number of intervals n.
    2. Compute the step size $h = \frac{(b-a)}{n}$.
    3. Add first and last ordinates once to get sum1
    4. Multiply the weighted sum by $\frac{h}{2}$.
    5. Apply weight 2 to intermediate ordinates and add them to get sum2.
    6. Compute the values of the ordinates.
    7. Obtain the weighted sum = sum1+sum2.

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Solve the following system of equations using the Gauss–Seidel iterative method. Perform three iterations. [U] / [CO1]

    4x + y = 5

    x + 5y = 6

16. Use bisection method to find a root of $e^x - 5 = 0$. Do at least 3 iterations. [U] / [CO1]

17. Explain how an exponential curve $y = ae^{bx}$ can be fitted using the method of least squares. [U] / [CO1]

18. Compute the first derivative of f(x) at x=0, from the following tabular data using finite differences. [U] / [CO1]

x	0.0	0.2	0.4	0.6
y=f(x)	1.00	1.16	3.56	13.96

19. The current flowing through a circuit varies with time as given below. Apply Simpson's 1/3 rule to find the total charge flowing through the circuit in 4 seconds. [A] / [CO2]

Time $t(s)$	0	1	2	3	4
Current $I(s)$	2.0	2.5	3.2	3.8	4.0

20. Evaluate $\int_0^6 \frac{1}{1+x^2} dx$ using Simpson's 3/8 rule. [E] / [CO2]

21. [E] / [CO2]

Solve the diffusion equation $\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2}, \quad 0 \le x \le 1$

Given:

$T = \sin(\pi x)$ when $t = 0,\ 0 \le x \le 1$ and
T=0 at x=0 and x=1 for t>0

Using Schmidt formulae, find the numerical solution at t=0.002 if
$\Delta t = 0.002 \quad and\ \Delta x = 0.5.$

22. Find y(0.1) for $\frac{dy}{dx} = x + y^2$, y(0)=1, h=0.1, using Modified Euler's method. [A] / [CO2]

[5x6 = 30]